

SEQUENTIAL LOGIC DESIGN

Module.3

CDA 3103C

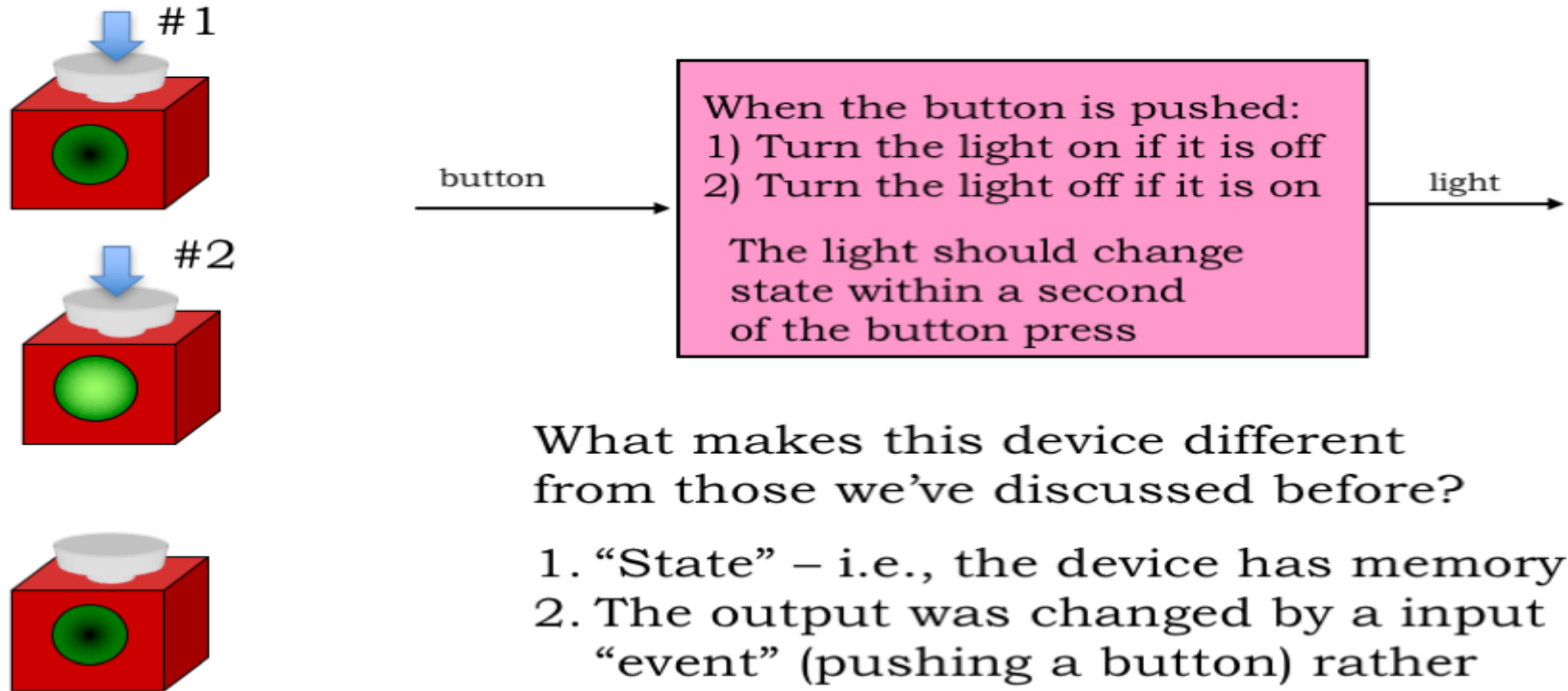
Mesut Ozdag, Ph.D.

Introduction

- Combinational Circuits
 - Without memory
 - Outputs depend only on current inputs
 - Output determined after sufficient time has elapsed
- Sequential Circuits
 - Have memory
 - Outputs depend on inputs and previous outputs
 - Previous output stored even after waiting

Something We Can't Build (Yet)

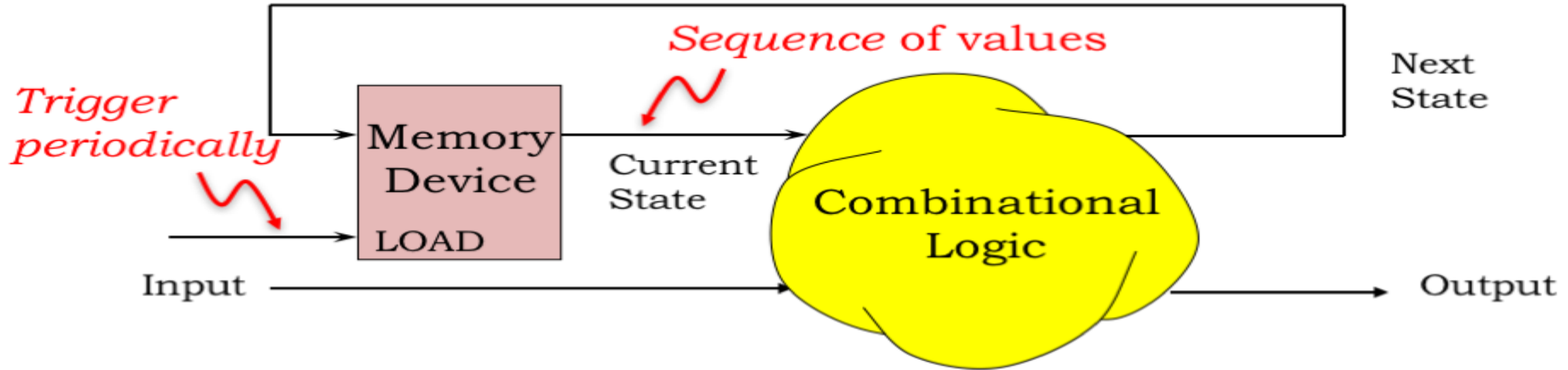
What if you were given the following design specification:



What makes this device different from those we've discussed before?

1. "State" – i.e., the device has memory
2. The output was changed by a input "event" (pushing a button) rather than an input "level"

Digital State: What We'd Like to Build



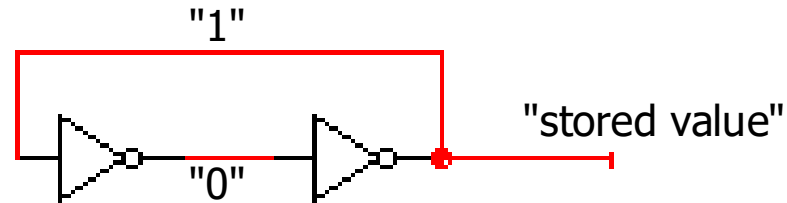
Plan: Build a Sequential Circuit with stored digital STATE –

- Memory stores CURRENT state, produced at output
- Combinational Logic computes
 - NEXT state (from input, current state)
 - OUTPUT bits (from input, current state)
- State changes on LOAD control input

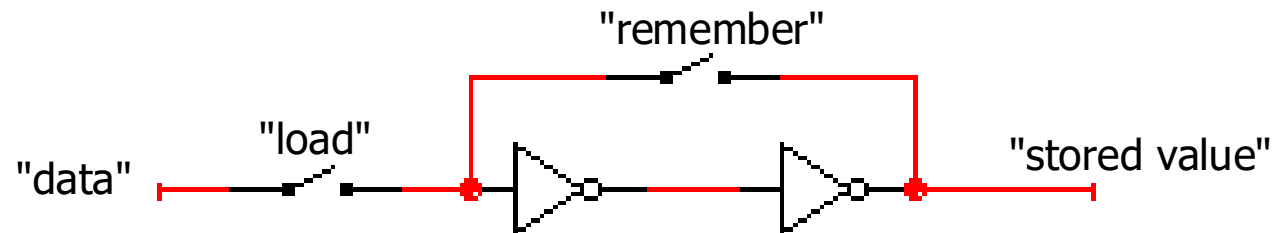
Needed:
Loadable
Memory

Simple Sequential Circuits

- Two inverters form a static memory cell
 - The previous output is stored by passing it back into the system
 - Will hold value as long as it has power applied



- How to get a new value into the memory cell?
 - Selectively break feedback path
 - Load new value into cell



Sequential Circuits

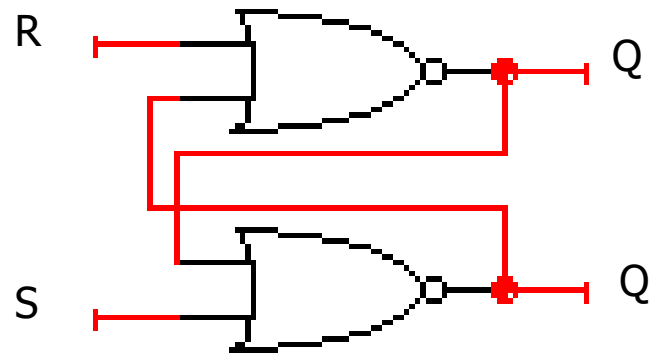
- Latches
- Flip-flops
- Registers

Latches

- A latch is a memory element in which the output is equal to the value of the stored state inside the element and the state is changed whenever the appropriate inputs change.

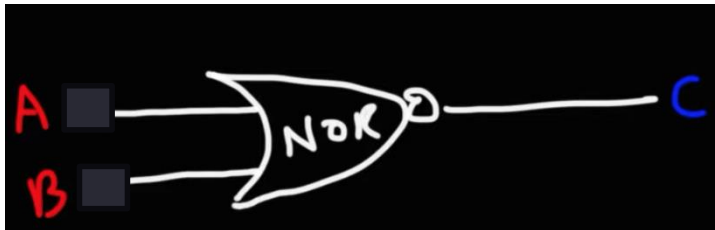
Set-Reset Latch

- Built from cross-coupled NOR gates
- Ability to force output to 0 (reset=1) or 1 (set=1)



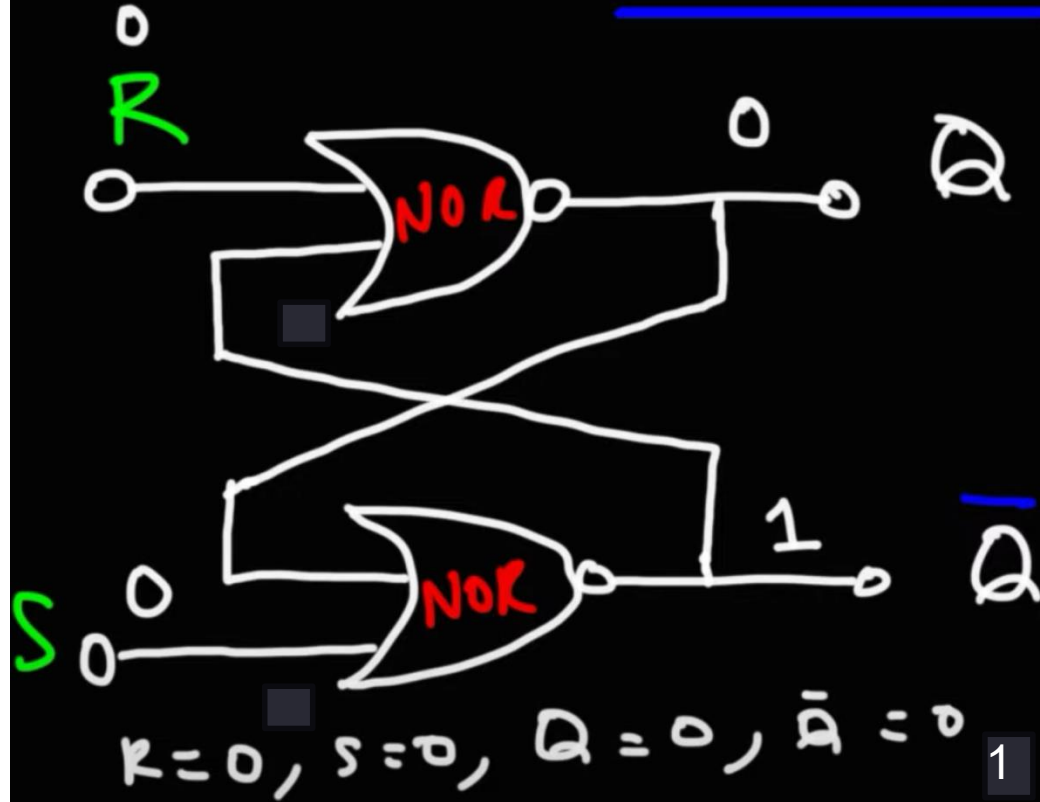
S	R	Output
0	0	No change
1	0	Set Q = 1
0	1	Reset Q = 0
1	1	Not Allowed

Note on nor gate:



A	B	C
0	0	1
1	0	0
0	1	0
1	1	0

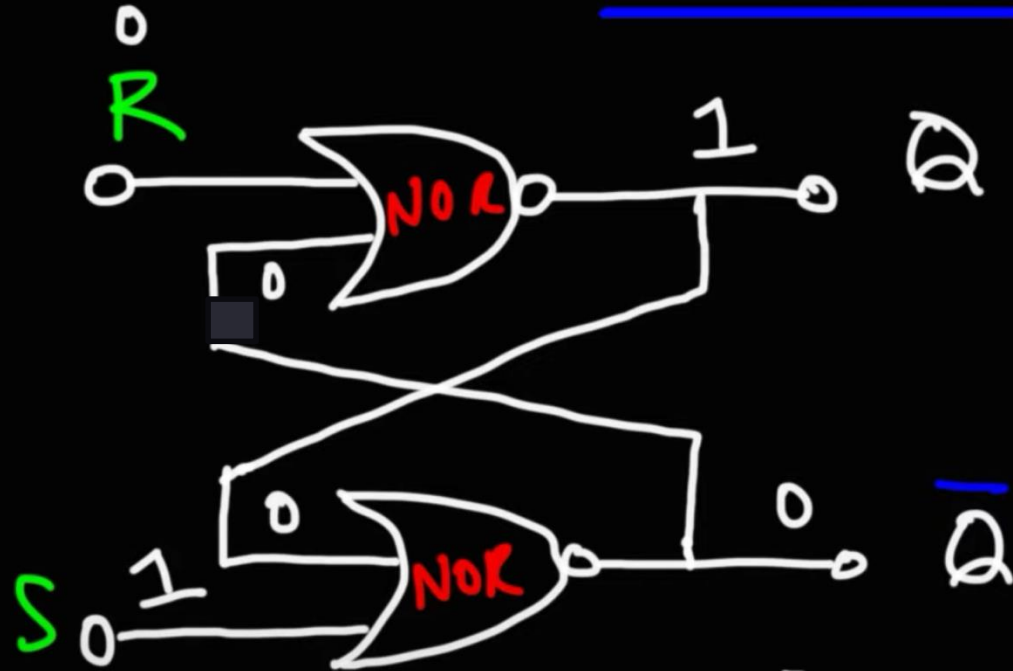
SR Latch



A	B	C
0	0	1
1	0	0
0	1	0
1	1	0

S	R	Output
0	0	No change
1	0	Set $Q=1$
0	1	Reset $Q=0$
1	1	Not Allowed

SR Latch



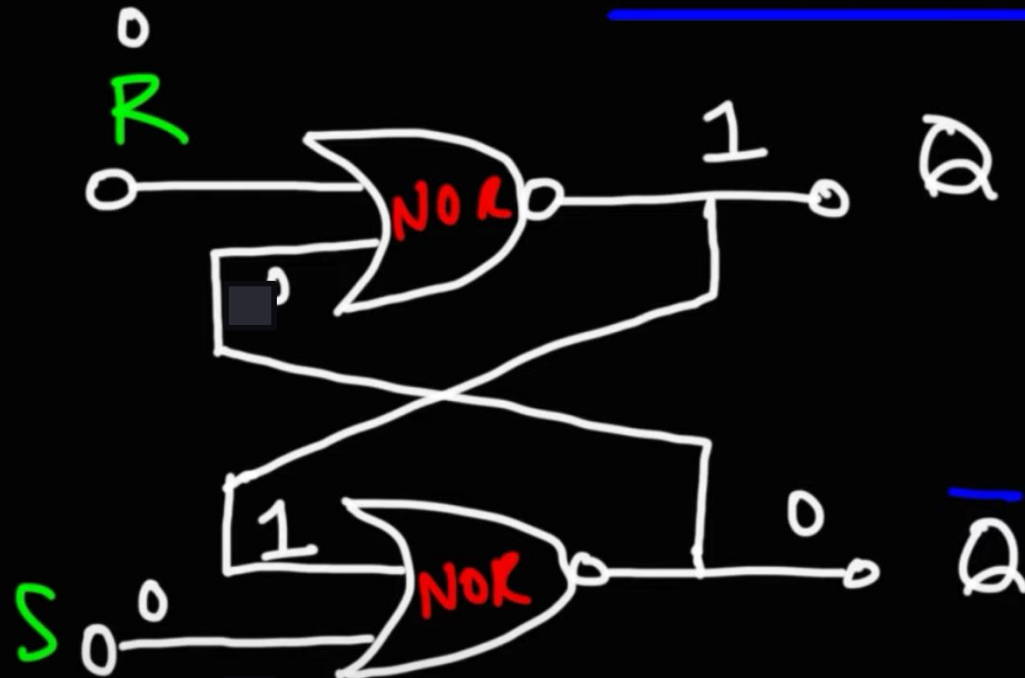
$R=0, S=0, Q=0, \bar{Q}=0$

$R=0, S=1, Q=1, \bar{Q}=0$

A	B	C
0	0	1
1	0	0
0	1	0
1	1	0

S	R	Output
0	0	No change
1	0	Set $Q=1$
0	1	Reset $Q=0$
1	1	Not Allowed

SR Latch

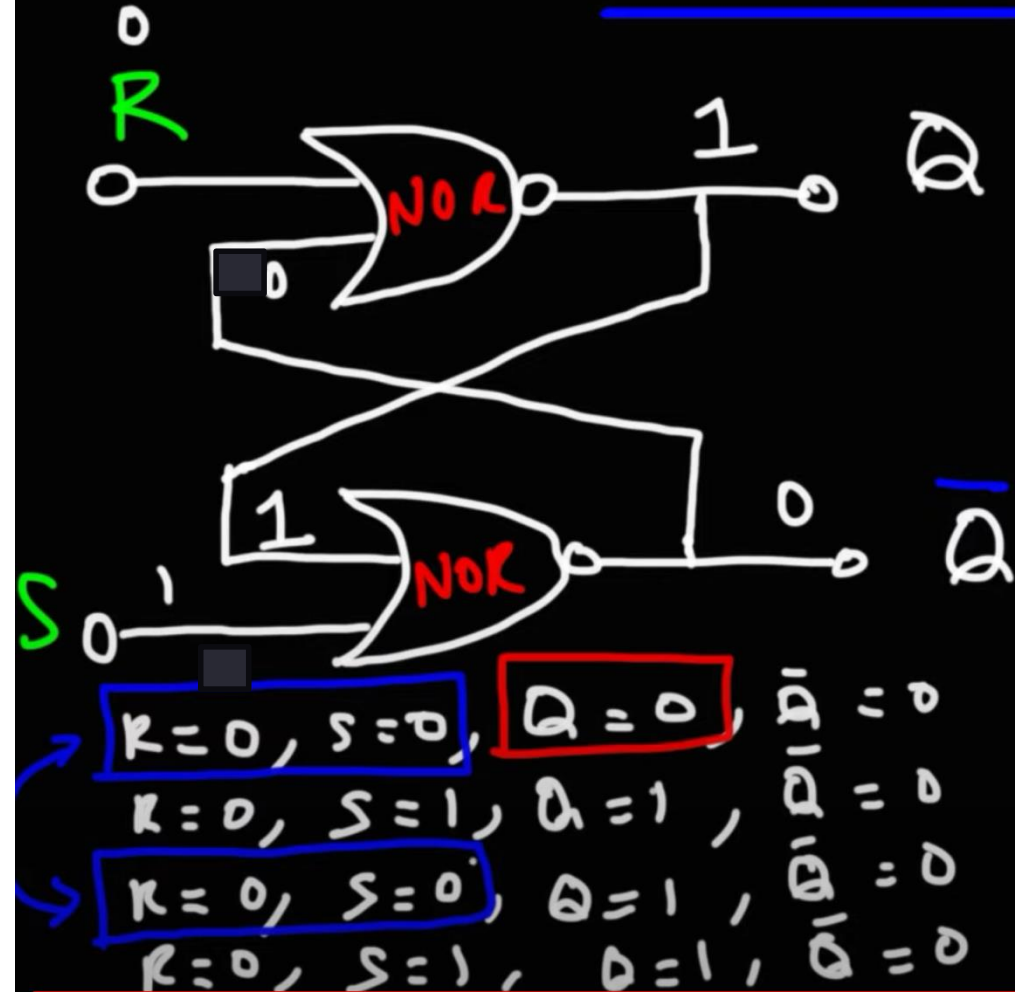


$R=0, S=0, Q=0, \bar{Q}=1$
 $R=0, S=1, Q=1, \bar{Q}=0$
 $R=0, S=0, Q=1, \bar{Q}=0$

A	B	C
0	0	1
1	0	0
0	1	0
1	1	0

S	R	Output
0	0	No change
1	0	Set $Q=1$
0	1	Reset $Q=0$
1	1	Not Allowed

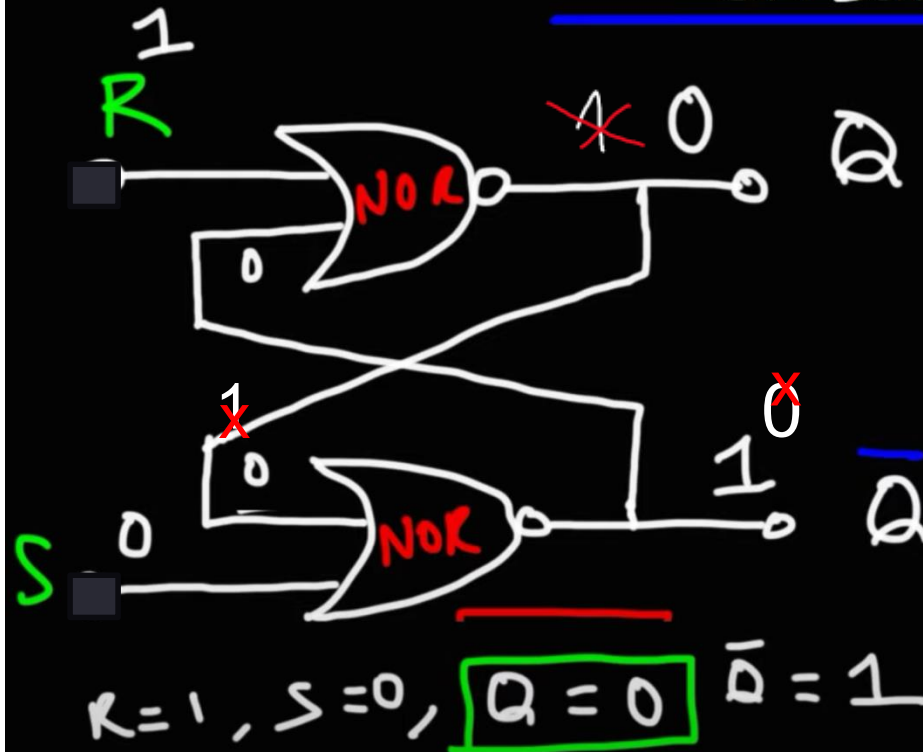
SR Latch



A	B	C
0	0	1
1	0	0
0	1	0
1	1	0

S	R	output
0	0	No change
1	0	Set $Q = 1$
0	1	Reset $Q = 0$
1	1	Not Allowed

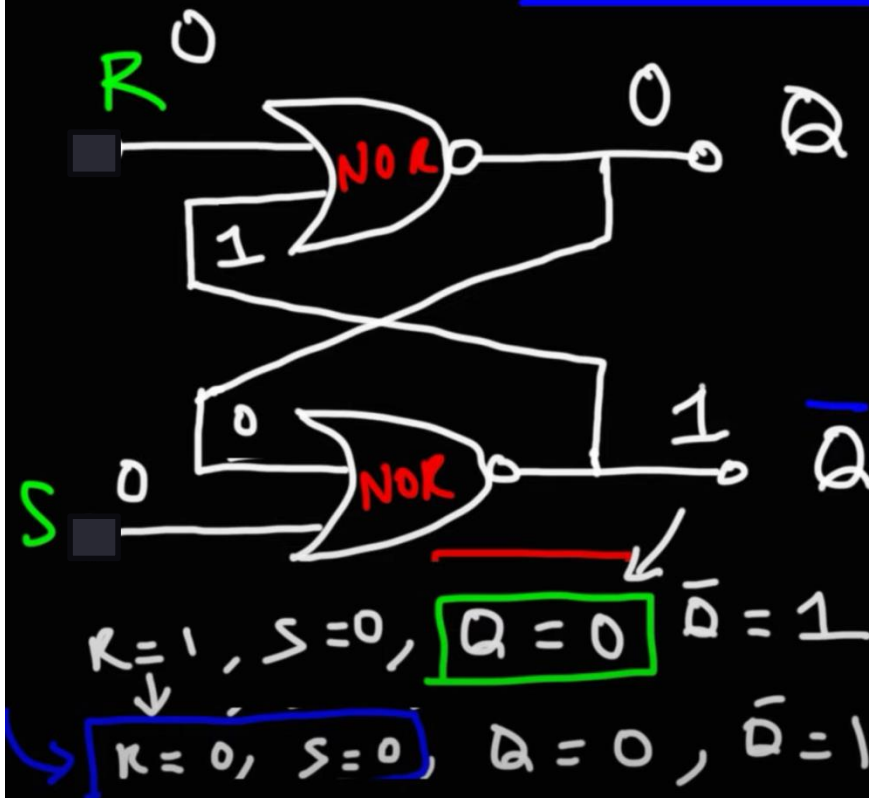
SR Latch



A	B	C
0	0	1
1	0	0
0	1	0
1	1	0

S	R	Output
0	0	No change
1	0	Set $Q = 1$
0	1	Reset $Q = 0$
1	1	Not Allowed

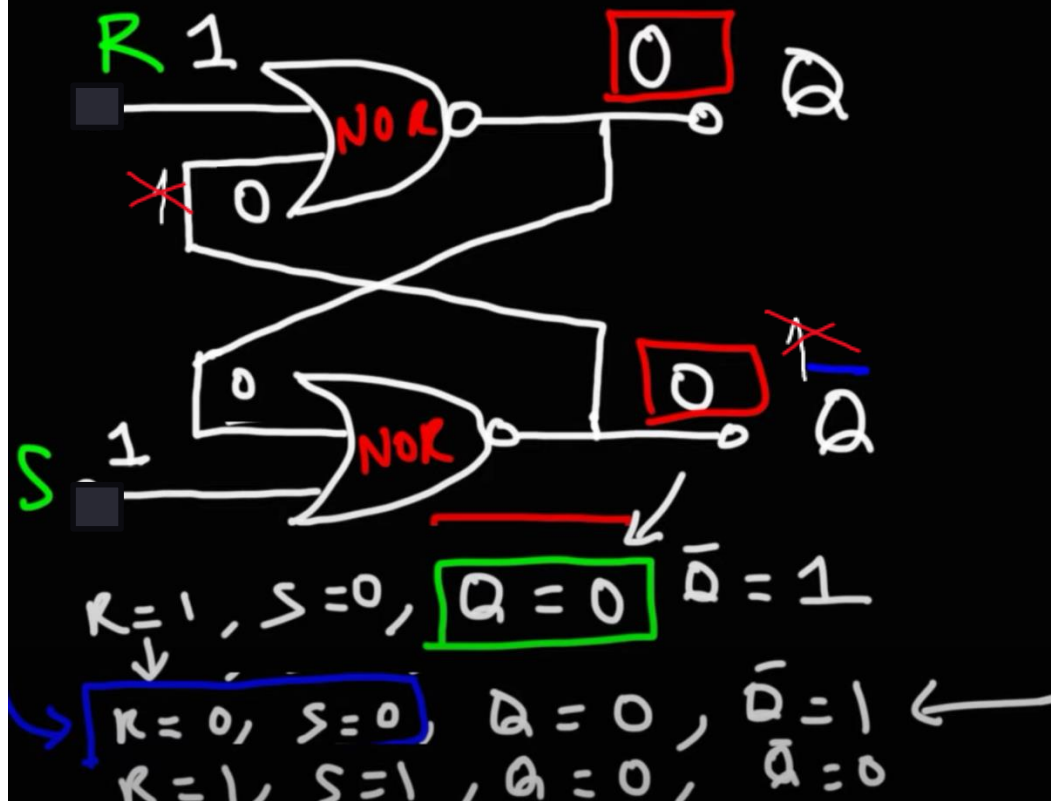
SR Latch



A	B	C
0	0	1
1	0	0
0	1	0
1	1	0

S	R	Output
0	0	No change
1	0	Set $Q = 1$
0	1	Reset $Q = 0$
1	1	Not Allowed

SR Latch



A	B	C
0	0	1
1	0	0
0	1	0
1	1	0

S	R	Output
0	0	No change
1	0	Set $Q = 1$
0	1	Reset $Q = 0$
1	1	Not Allowed

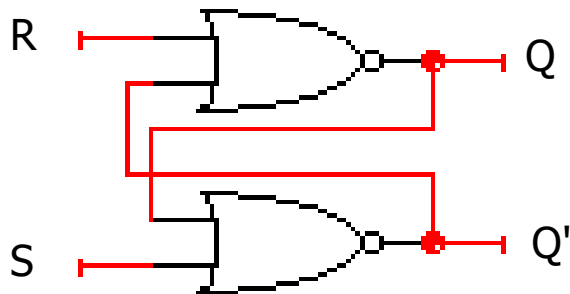
State Machines

- A sequential circuit is described as a finite-state machine
 - A set of states
 - A function to determine the next state
 - A function to determine the output
 - Functions can still be represented by a truth table

Behavior of the S-R Latch

- States: {hold, set, reset, unstable}
- Next-State function:

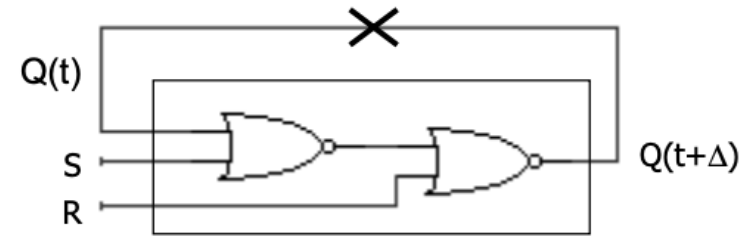
S	R	State
0	0	hold
0	1	0
1	0	1
1	1	unstable



Behavior of the S-R Latch

- States: {hold, set, reset, unstable}
- Output function:

S	R	Q(t)	Q(t+ Δ)
0	0	0	0 (hold)
0	0	1	1 (hold)
0	1	0	0 (reset)
0	1	1	0 (reset)
1	0	0	1 (set)
1	0	1	1 (set)
1	1	0	? (not allowed)
1	1	1	? (not allowed)



		S	
	0	0	1
Q(t)	0	X	1
	1	X	1
		R	

characteristic equation
 $Q(t+\Delta) = S + R' Q(t)$



MESUT OZDAG, PH.D.
MESUT.OZDAG@UCF.EDU

